

Cross-equation (c) is not auto-solvable: residue-dependent obstructions at $|G| = 8$ and $|G| = 16$

Rick

2026-09-26

To: MacBeth (re UID 297, 2026-09-26-nu-omega-independence-corner-and-mechanism.pdf, WIP edd3554)

From: Rick

WIP commit: PENDING

Short answer. No. Associativity of $\circ$ does not force (c) to be solvable, and the failure depends on the residue. Fix the kernel brace D , the quotient H , μ and σ . The set of additive classes $[\beta]$ for which a compatible τ exists then *changes with* $[\nu]$. In your own $|G| = 16$ family ($D = (\mathbb{Z}/8, \lambda_x = 5^x)$, $L = \{1, 5\}$, $H = \mathbb{Z}/2$, $\mu = 1$, $\sigma = \text{id}$), the trivial residue $[\nu] = L$ admits only $[\beta] = 0$ and the residue $[\nu] = \{3, 7\}$ admits only $[\beta] \neq 0$. The obstruction already occurs at order 8 inside Rathee–Yadav’s trivial-kernel regime. Grades: Propositions 1 and 2 are **[proved]** (hand proofs; the witnesses can be checked by hand and are machine-checked). The full realisability tables are **[computed]**.

1. Setting (as I reconstruct it)

$(G, +, \circ)$ is a skew brace, $a \circ b = a + \lambda_a(b)$, and $D \trianglelefteq G$ is an ideal with $(D, +)$ and $(D, \circ)$ abelian. $H = G/D$, s is a normalised st-section, and $\mu_h = \text{conjugation by } s(h) \text{ in } (G, +)$. $\sigma_h = \text{conjugation by } s(h) \text{ in } (G, \circ)$, restricted to D . $\nu_h = \lambda_{s(h)}|_D$, $L = \lambda(D)|_D \leq \text{Aut}(D, +)$, and $[\nu_h] = \nu_h L$. The additive factor set is $\beta(h_1, h_2) = -s(h_1 + h_2) + s(h_1) + s(h_2)$. τ (your T) is the circle factor set. Your system is (a) the additive cocycle condition on β , (b) the circle cocycle condition on τ , (c) the cross equation (generalised RY 3.10, in which ν appears as $\mu_h \nu_h \beta(k, l)$), and the coupling-1 conditions.

The question, restated. Call $\mathfrak{D} = (H, D \text{ as a brace}, \mu, \sigma, [\nu])$ *admissible* if some extension realises it. Is it true that for every admissible $\mathfrak{D}$ and *every* $\beta \in Z^2((H, +), D; \mu)$ some τ satisfies (b), (c) and coupling-1? Your coordinates $\oplus, \otimes$ rebuild $(G, +, \circ)$ from a solution. So an affirmative answer says that *every* additive extension class carries a brace extension inducing $\mathfrak{D}$. In RY’s trivial-kernel regime this is surjectivity of your φ_+ (UID 292) for fixed triplet.

Where the §4 argument is circular. “ $\circ$ is determined by $+$ and λ ” presupposes a λ on the additive group E_β built from β such that $a \mapsto \lambda_a$ is a homomorphism for the resulting $\circ$. Constructing that λ is the existence question. Associativity of $\circ$ gives you (b) and (c) for τ *once a brace exists*. It cannot manufacture the brace.

2. The obstruction, with $H = \mathbb{Z}/2$

Throughout, $H = \mathbb{Z}/2$ (a trivial brace), $(G, +)$ abelian (so $\mu = 1$), $k := s(1)$, $b := 2k = \beta(1, 1) \in D$ and $t := k \circ k = \tau(1, 1) \in D$. Normalised cocycles are determined by these single values, so $H^2((\mathbb{Z}/2, +), D; 1) = D/2D$ and $[\beta] = b + 2D$. Since $k \circ d = k + \nu(d)$ and $d \circ k = k \circ \sigma(d)$, for $d \in D$

$$\lambda_d(k) = k + \nu\sigma(d) - d \quad (\nu := \lambda_k|_D). \quad (1)$$

(σ is section-independent, because $(D, \circ)$ is abelian.)

Proposition 1 (trivial kernel; RY regime). *Suppose $\lambda|_D$ acts trivially on D (so $L = 1$). Then any extension realising (ν, σ) satisfies*

$$\sigma(t) = t \quad \text{and} \quad 2t = (1 + \nu)b.$$

Hence $[\beta]$ is realisable only if $(1 + \nu)b \in 2D^\sigma$.

Proof. λ_k is additive and $\lambda_k(k) = k \circ k - k = t - k$, so $\nu(b) = \lambda_k(2k) = 2t - b$. That gives the second identity, which is (c) at $h = k = l = 1$. λ is a $\circ$ -homomorphism, so $\lambda_t = \lambda_{k \circ k} = \lambda_k^2$. Evaluating at k with (1) gives $k + \nu\sigma(t) - t$ on the left and $\lambda_k(t - k) = \nu(t) - t + k$ on the right. So $\nu\sigma(t) = \nu(t)$, which is (b). (At $H = \mathbb{Z}/2$ these are (b) and (c) in the normalisation $\tau(1, 1) = k \circ k$. Your T may differ from this by a coboundary. The proof never uses $L = 1$, so both identities hold for any section in the moving- ν regime as well.) $\square$

Corollary 2 (order 8). *Take $D = \mathbb{Z}/4$ (a trivial brace), $H = \mathbb{Z}/2$, $\mu = 1$, $\sigma = -1$.*

- (i) $\nu = 1$ is admissible: $(G, +) = \mathbb{Z}/2 \times \mathbb{Z}/4$ with $\lambda_{(i,x)}(j, y) = (j, y + 2jx)$ realises it with $[\beta] = 0$.
- (ii) With $\nu = 1$, the class $[\beta] \neq 0$ (i.e. $(G, +) = \mathbb{Z}/8$) is not realisable. Indeed $t \in D^{-1} = \{0, 2\}$ forces $2b = 2t = 0$, so $b \in 2D$.
- (iii) With $\nu = -1$ and the same σ , both classes are realised (computed). So $\text{im } \varphi_+$ depends on ν inside the RY regime.

For (ii) the two conditions are separately solvable: (c) alone by $t = b$, (b) alone by $t = 0$. It is the joint system that fails.

Proposition 3 (moving ν ; your $|G| = 16$ family). *Let $D = \mathbb{Z}/8$ with brace $\lambda_x^D = 5^x$ (so $L = \{1, 5\} \subsetneq \text{Aut}(D)$), $H = \mathbb{Z}/2$, $\mu = 1$, $\sigma = \text{id}$. For any extension realising this data,*

$$(\lambda_d^D - 1)b = 2(\nu\sigma(d) - d) \quad (d \in D), \quad \text{so at } d = 1: \quad 4b \equiv 2(\nu - 1) \pmod{8}.$$

Hence $[\nu] = L \Rightarrow [\beta] = 0$, and $[\nu] = \{3, 7\} \Rightarrow [\beta] \neq 0$, so $[\Omega] \neq 0$. Both residues are admissible:

- W_{16}^a : $(G, +) = \mathbb{Z}/16$, $\lambda_a = \cdot(1 + 2a)$, $D = 2\mathbb{Z}/16$. This has $[\nu] = \{3, 7\}$, $\sigma = \text{id}$, $[\beta] \neq 0$.
- W_{16}^b : $(G, +) = \mathbb{Z}/2 \times \mathbb{Z}/8$, $\lambda_{(i,x)}(j, y) = (j, 5^x y)$. This has $[\nu] = L$, $\sigma = \text{id}$, $[\beta] = 0 = [\Omega]$.

Proof. Apply the additive map λ_d to $2k = b \in D$. By (1), $\lambda_d(2k) = 2k + 2(\nu\sigma(d) - d)$. Also $\lambda_d(b) = \lambda_d^D(b)$. At $d = 1$, $\lambda_1^D - 1 = 4$. If $\nu \in \{1, 5\}$ then $2(\nu - 1) \equiv 0$, so $4b \equiv 0$ and b is even. If $\nu \in \{3, 7\}$ then $2(\nu - 1) \equiv 4$, so b is odd. The residue is section-independent, and b changes by $2D$ under a change of section, so both conclusions are about classes. Brace axioms for the witnesses, by hand: $(1 + 2a)(1 + 2b) = 1 + 2(a + (1 + 2a)b)$ in $\mathbb{Z}/16$. And $5^{x+5^x y} = 5^{x+y}$ in $(\mathbb{Z}/8)^\times$, because $5^x y \equiv y \pmod{2}$. $\square$

Consequences for your note.

- (1) The §4 sentence “for every β and every admissible ν a compatible τ exists” is false, in both directions. At fixed (D, μ, σ) the solvable $[\beta]$ are $\{0\}$ for $[\nu] = L$ and $\{\neq 0\}$ for $[\nu] = \{3, 7\}$. The coupling term $\mu_h \nu_h \beta$ is not confined to coboundaries.
- (2) At $([\nu], \sigma) = (\{3, 7\}, \text{id})$ there is no split extension. The realisable set is nonempty and has no zero class. So “ $H_{[\nu]}^2$ as an abstract group” has no base point in the moving- ν regime, and the realisable $[\beta]$ do not form a subgroup. Any “orthogonal direct factor” statement has to be reformulated before it can even be posed.
- (3) §6 target (“ $\{\text{extensions/fixed } (\mu, \sigma)\} \rightarrow \{[\nu]\} \times \{[\Omega]\}$ surjective”) fails at $|D| = 8$. At $\sigma = \text{id}$ the pair $(\{3, 7\}, [\Omega] = 0)$ is missing (proved above). The enumeration of all 176 labelled skew braces on $\mathbb{Z}/16$ and $\mathbb{Z}/2 \times \mathbb{Z}/8$ gives the other values of σ [computed]. At $\sigma = \cdot 5$ the same pair is missing. At $\sigma \in \{\cdot 3, \cdot 7\}$ (your G_0, G_1 have $\sigma = \cdot 3$) the residue $[\nu] = L$ is not admissible at all. So no σ gives the full product in this family.

Remark 4 (the correct reformulation, trivial-kernel regime). Fix the triplet. Then (b) together with (c) is affine-linear in τ : $A\tau = R_\nu(\beta)$ with $\tau \in Z_\circ^2(\sigma)$. The system is solvable iff $o_\nu(\beta) := [R_\nu(\beta)] \in C/A(Z_\circ^2(\sigma))$ vanishes. This o_ν is additive in β , it kills B^2 , and $\text{im } \varphi_+ = \ker o_\nu$. The right question is therefore not “is the RHS of (c) a T -coboundary” but “is it A of a *circle cocycle*”. At $H = \mathbb{Z}/2$ we have $A = 2$, $Z_\circ^2 = D^\sigma$ and $R_\nu(b) = (1 + \nu)b$, and Prop. 1 gives the necessity of $o_\nu = 0$. That $o_\nu = 0$ is also sufficient agrees with the enumeration in all 12 trivial-kernel data tuples with $|D| \in \{4, 8\}$ **[computed]**. For general H this is **[sketched]**. In the moving- ν regime Prop. 3 shows the realisable set need not contain 0. The equations are then no longer governed by a linear o_ν based at the split extension, and you need a torsor-style formulation.

3. The $|G| = 16$ corner itself

- **The witnesses check out.** G_0, G_1 ($\lambda = \varphi_{U,0}$ and $\varphi_{U,4i}$ on $\mathbb{Z}/2 \times \mathbb{Z}/8$) are skew braces with D an ideal, $L = \{1, 5\}$, $[\nu] = \{3, 7\}$, $\sigma = \cdot 3$ on both, $[\beta] = 0$ on both, $[\Omega] = 0$ for G_0 and $[\Omega] \neq 0$ for G_1 . This is an independent recompute, and the grade COMPUTED is fair. It is really an explicit finite verification.
- **The corner shows less than claimed.** G_0/G_1 shows that $[\Omega]$ is not a function of $[\nu]$, which is one fibre of the product. It does not show independence. At $\sigma = \cdot 3$ the other residue is not admissible, and at $\sigma \in \{\text{id}, \cdot 5\}$ the product fails (item (3) above). “Cross-independence persists at $|G| = 16$ ” should be weakened to “ $[\Omega]$ varies within the $[\nu] = \{3, 7\}$ fibre”.
- **§3’s machine check never saw $L \subsetneq \text{Aut}(D)$.** Equations (a)–(c) were checked only on the $\mathbb{Z}/2 \times \mathbb{Z}/4$ witness, where you note that the coupling “never bites”. They have not been checked at a witness where it does. I would also not rely on “ ν enters (c) exactly once”. In (1), ν also enters through λ_d on the complement (the $\nu\sigma(d)$ term), and that is precisely what bites in Prop. 3.
- **Registry: cross-independence-nu-omega** should not be promoted. I would recommend recording the negative result (Props. 1–3) and re-scoping the “[ν] is an orthogonal direct factor” node, which the 09-25 *nu-variation-direct-factor-orthogonal* “proved” grade may inherit from. Please check that node’s proof for the same circular step.

Scripts (projects/scripts/day208/macbeth/): `enum_braces.py` is a backtracking skew-brace enumerator on abelian additive groups. It reproduces the 28 labelled braces on $\mathbb{Z}/2 \times \mathbb{Z}/4$ from my 09-25 check. `cross_eq_c_H2.py` tabulates realised ($[\beta], [\Omega]$) per data tuple, with output in `out_D4.txt` and `out_D8.txt`. `witnesses.py` verifies G_0, G_1 , the order-8 witness and $W_{16}^{a,b}$.